# LOCO PILOT POCKET BOOK

GE Transportation
a Wabtec company





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# WDG4G - 4500 HP

Specification

| | Description | Parameters |
|---|---|---|
| General | Wheel arrangement | CO-CO |
| | Weight (kgs) | 132,000 |
| | HP | 4500 |
| | HP for Traction | 4261 |
| | Loco Dimension (L x W x H) in mm. | 22573 x 3200 x 4227 |
| | New wheel diameter (mm) | 1092 |
| | Condemn wheel diameter (mm) | 1016 |
| Engine | Model | GEVO 12 |
| | No of cylinders | 12 |
| | Maximum RPM | 1050 |
| | Cycle | 4 |
| | Fuel tank capacity – usable oil (Liters) | 6000 |
| | Lubricating oil capacity (Liters) | 1030 |
| | Coolant capacity (Liters) | 1249 |
| | Turbo charger | Single stage - one no. |

| | Description | Parameters |
|---|---|---|
| Propulsion | # of Traction inverters | 6 |
| | # of Traction motors – AC Induction | 6 |
| | Gear ratio | 85:18 |
| | Maximum speed (kmph) | 100 |
| | Maximum starting tractive effort (kN) | 544 |
| | Max. continuous tractive effort (kN) | 405 |
| | Max. braking effort (kN) | 272 |
| | | |
| | | |
| | | |
| Others | Sand in Liters | 340 |
| | # of Sand boxes – 85 Liters each | 4 |
| | Type of air brake system → Fast brake / Electronic Air Brake (EAB) | |
| | | |
| | | |

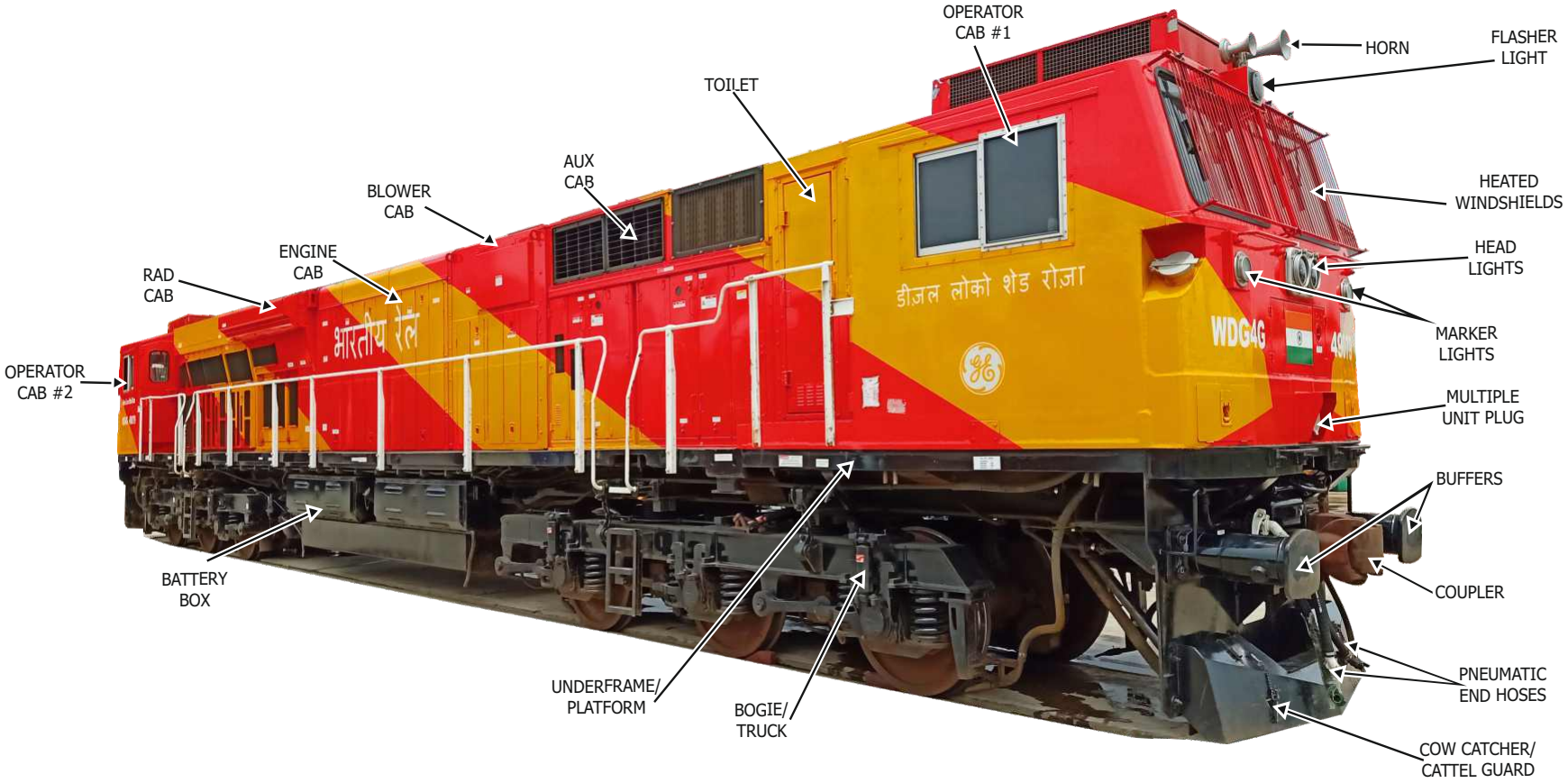
# WDG6G - 6000 HP

Specification

| | Description | Parameters |
|---|---|---|
| General | Wheel arrangement | CO-CO |
| | Weight (kgs) | 138,000 |
| | HP | 6000 |
| | HP for Traction | 5686 |
| | Loco Dimension (L x W x H) in mm. | 22319 x 3250 x 4227 |
| | New wheel diameter (mm) | 1092 |
| | Condemn wheel diameter (mm) | 1016 |
| Engine | Model | GEVO 16 |
| | No of cylinders | 16 |
| | Maximum RPM | 1050 |
| | Cycle | 4 |
| | Fuel tank capacity – usable oil (Liters) | 8000 |
| | Lubricating oil capacity (Liters) | 1300 |
| | Coolant capacity (Liters) | 1468 |
| | Turbo charger | Single stage – 2 nos. |

| | Description | Parameters |
|---|---|---|
| Propulsion | # of Traction inverters | 6 |
| | # of Traction motors – AC Induction | 6 |
| | Gear ratio | 85:16 |
| | Maximum speed (kmph) | 100 |
| | Maximum starting tractive effort (kN) | 570 |
| | Max. continuous tractive effort (kN) | 420 |
| | Max. braking effort (kN) | 285 |
| | | |
| | | |
| Others | | |
| | Sand in Liters | 630 |
| | # of Sand boxes – 8 | 2x90 lts. & 6x75 lts. |
| | Type of air brake system → Fast brake / Electronic Air Brake (EAB) | |
| | | |
| | | |

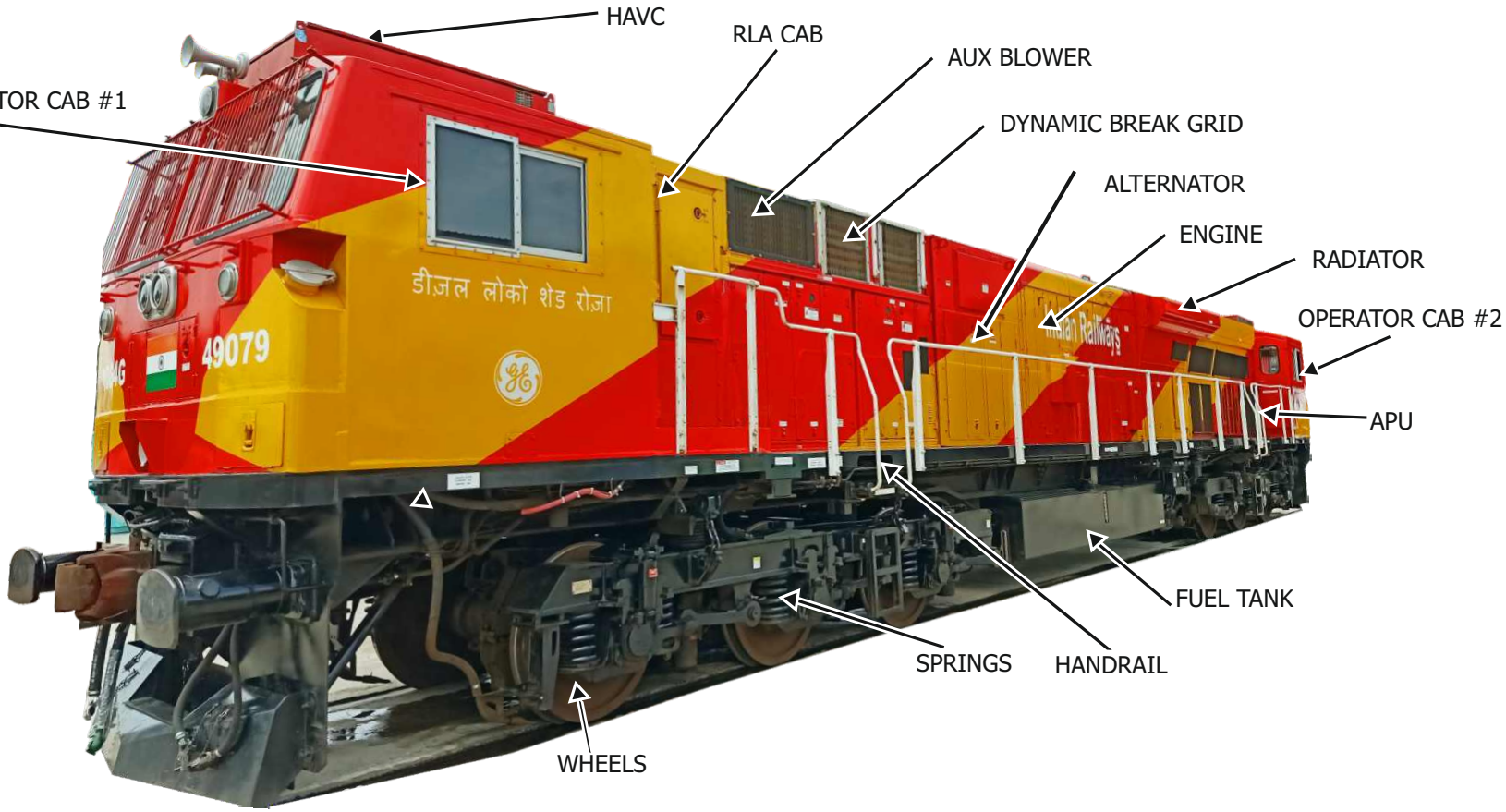
# WDG4G - 4500 HP




## A-SIDE

Locomotive Layout

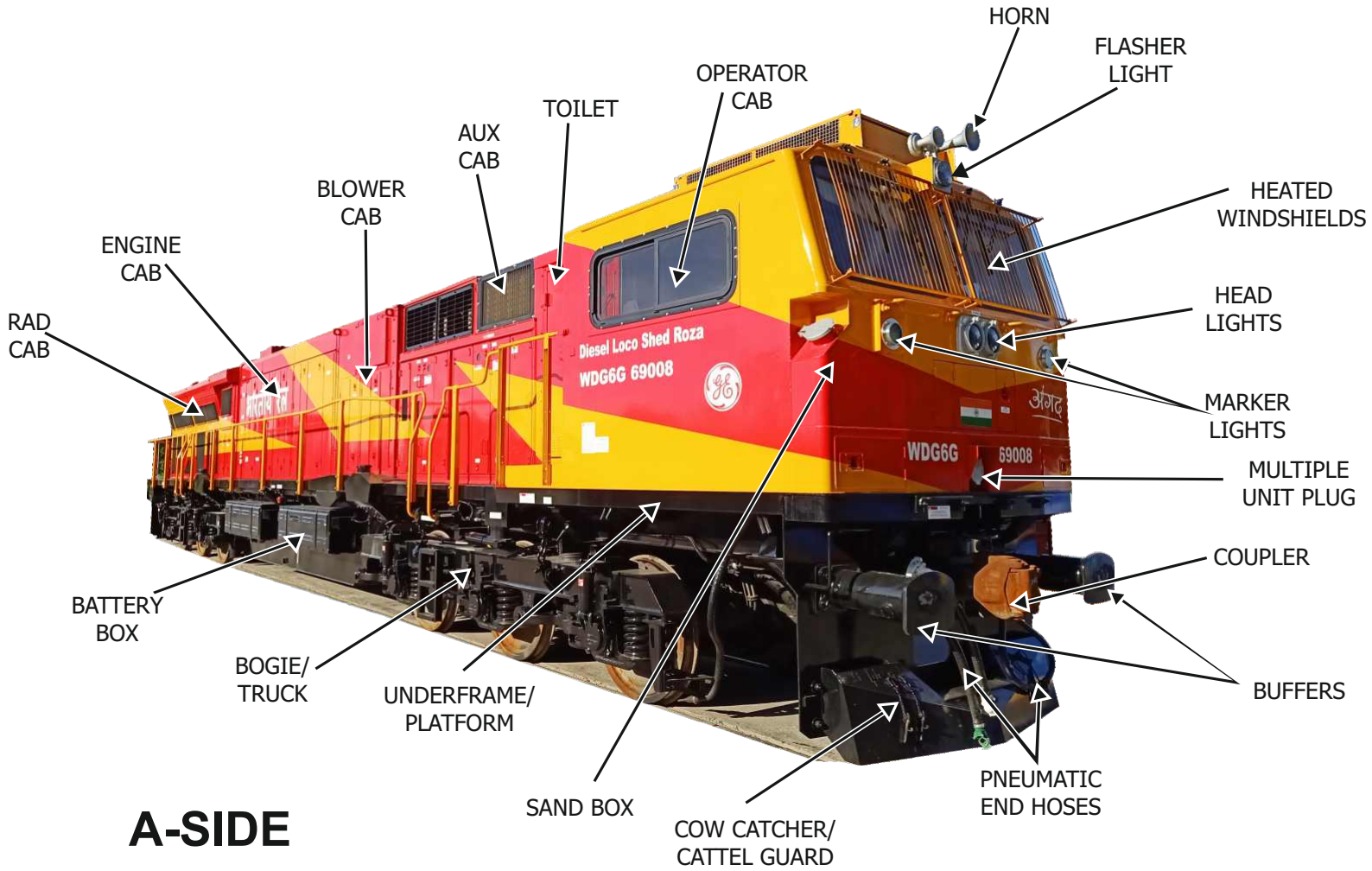
# WDG4G - 4500 HP




Locomotive Layout

## B-SIDE

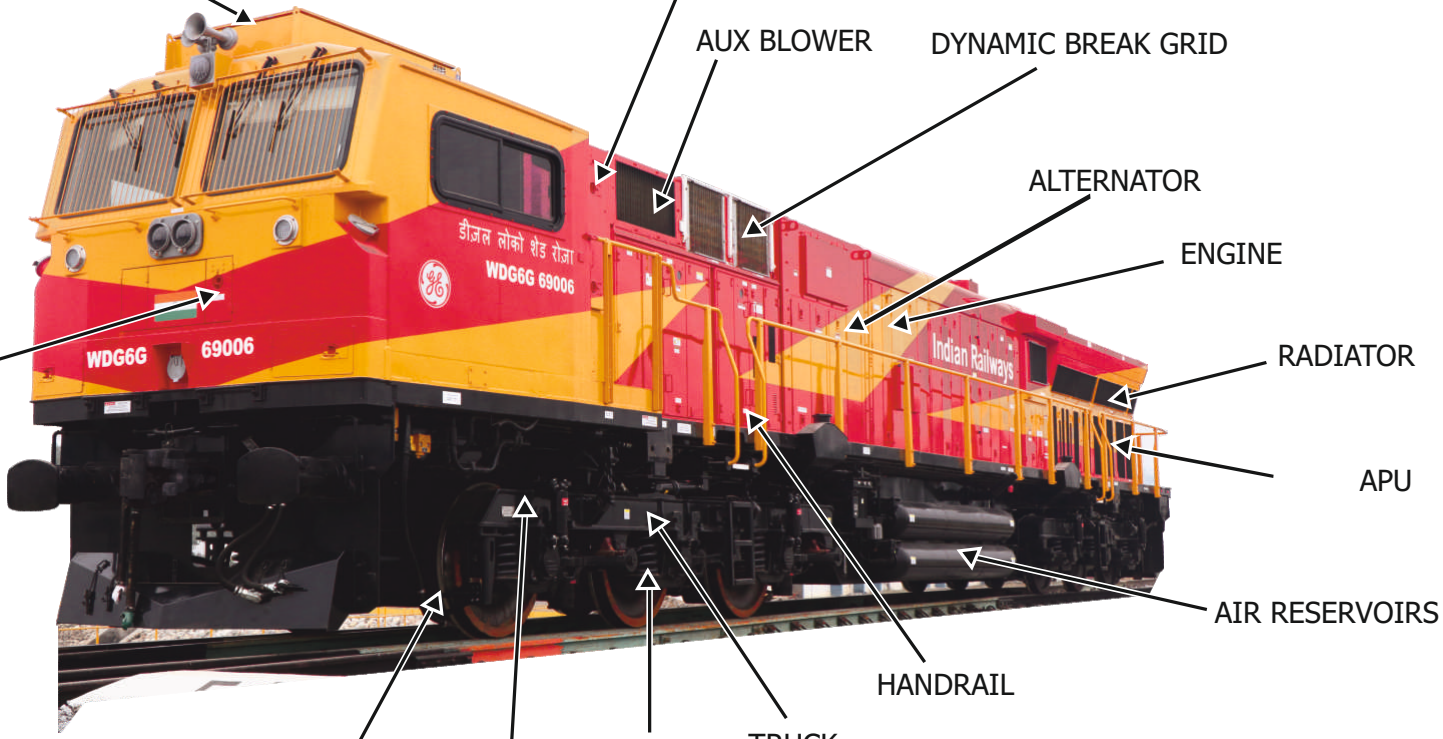
# WDG6G - 6000 HP




**A-SIDE**

Locomotive Layout

# WDG6G - 6000 HP




**B-SIDE**

Locomotive Layout

## BEFORE BOARDING LOCOMOTIVE

Follow the below steps before boarding the locomotive:

1. Inspect for broken, worn, loose or dragging parts (brake rigging, brake shoes, wheels, covers, etc.)
2. Check for leaks from outside piping.
3. Properly position all drain and air cutout cocks.
4. Ensure the Brake Pipe cocks are properly positioned.
5. Check for proper connections of the air hoses and MU jumper cables (if unit is in a consist)
6. Check the fuel level on the fuel sight glass on the platform on both A and B sides of the locomotive.
7. Ensure the Dead engine/Dead (in tow) cutout cock on the Pneumatic Operating Unit (POU) is in the correct position on each locomotive.

## AFTER BOARDING LOCOMOTIVE

Follow the below steps after boarding the locomotive:

1. With the engine shutdown, remove rags, tools etc. from moving parts and electrical equipment of all consist locomotives.
2. Ensure that the exhaust stack cover has been removed (if applied).
3. Check the diesel Engine lubricating-oil supply in all consist locomotives. Oil level should be above ADD mark on the dipstick with the engine at idle. The dipstick is located on the side of the engine.
4. Check the air compressor lubricating-oil level.
5. Check the cooling water supply. Ensure the water drain valve is close.
6. Ensure the MR1 sensor cutout cock on the CA12 box is open.
7. Ensure the air brakes are set up properly.
8. Move the Engine Control (EC) switch to ISOLATE.
9. Check that the Master Controller handle is in ‘Idle’ and the Reverser handle is in the ‘Center’ position.



# Pre Checks



## Compressor oil




| Location | Rad cab |
|---|---|
| Check Points | Compressor oil should be in between 'Min' (above bottom red colour) and 'Max' mark when seen through spy glass |
| Note | To be checked 15 min after Loco shutdown |

## Cooling water for Engine



| Location | Rad cab & Engine cab junction |
|---|---|
| Check Points | Cooling water should be above 'Minimum' mark when seen through spy glass |
| Note | To be checked during Loco is running in 'Idle' condition |

## Engine Lube Oil




| Location | Engine cab – Both sides of Loco |
|---|---|
| Check Points | Old dip stick: At 'ADD' mark locos were allowed to run for another 400 KM or 02 day of operation.<br>New dip stick: At 'ADD' Mark Locos are not allowed to run & lube oil required to be topped up |
| Note | To be checked during Loco is running in 'Idle' condition |

# Engine Cranking



**Note:** Before cranking, ensure all breakers and EC switch in Cab #2 are in off condition

| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Close Battery Switch (BS) | Below EC panel | Cab #1 |
| 02 | Close all Circuit breakers | EC panel / ALP side front panel | Breaker Panel:<br>• 3rd & 4th row → All breakers – Left to right<br>• 1st & 2nd row → As required<br>ALP panel:<br>• Keep all 3 breakers on<br>• 'MU Shut down' switch should be released |
| 03 | Move the EC switch from 'Off' to 'Isolate' | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 04 | Ensure Throttle in 'Idle' & Reverser in 'Center' | Working cab | |
| 05 | Push the Engine start push button | EC panel | Hold the button in pushed condition until 'Engine start in progress' message appears on the screen |
| 06 | Engine picks up the speed and start running at idle speed → Observe on the screen | Working cab | Listen Crank warning bell & Engine pick up noise during the operation |
| 07 | Aux equipment's start functioning → Observe on the screen | Working cab | Battery charging, Compressor (MR pressure) etc. (Acknowledge any penalties or faults) |
| 08 | Check Engine Lube Oil level | Engine cab | By 'Dip stick' check method |

# Loco Running



| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Move the EC switch from 'Isolate' to 'Run' | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 02 | Ensure 'GFB' & ' Dynamic' breakers are ON | ALP panel / EC panel in cab-2 | |
| 03 | Release the Parking brake – If applied | EC panel | 'P' marker on TCD should not be displayed |
| 04 | Move → SA9 → Release, A9 → Run | Working cab | BC = 0 kg/cm², BP=5 kg/cm² |
| 05 | Put Reverser → 'Forward / Reverse' → Move the Throttle from 'Idle' to N1, N2… | Working cab | Watch the increase in kN on the screen. Allow 2 to 3 Seconds while moving from one notch to another notch. |



# Engine Shutdown



| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Move the Throttle handle to 'Idle' | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 02 | Apply SA9 | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 03 | Ensure Loco is Stopped, Keep Reverser → Center | | |
| 04 | Check on TCD screen for Throttle and Reverser status | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 05 | Switch off 'GFB' breaker | EC panel | |
| 06 | Move the EC switch from 'Run' to 'Isolate' | Working cab | Either cab #1 or cab #2 based on the cab selection |
| 07 | Allow the Engine to run for 5 min | | To stabilize all Engine sub systems |
| 08 | Apply Parking brake | Working cab | |
| 09 | Press 'Stop' push button | EC panel | Red colour |
| 10 | If it is MU → Press 'MU shutdown' for rear loco engine shutdown | Working cab, Opp of ALP desk | Either cab #1 or cab #2 based on the cab selection |

# Loco Shutdown



| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Switch off all the breakers | EC panel | Top to bottom → Right to Left |
| 02 | Move EC switch from 'Isolate' to 'Off' | EC panel | Both the cabs as applicable |
| 03 | Open the BS switch | Below EC panel | Battery will drain out if BS switch is not opened |
| 04 | Secure both the cabs – Close all the windows and doors | Both the cabs | Before leaving the Loco |



# Cab Changing-Single Loco



Situation:
Loco pilot is driving the Loco from Cab #1 and Cab #2 is inactive → Loco pilot wants to drive the loco from Cab #2 and make Cab #1 inactive.

Follow the below steps for cab changeover

## Step 1 (Cab #1)

- Throttle → Idle
- SA9 → Apply
- Reverser → Center & Remove
- Parking brake → Apply (Optional)
- EC Switch → Off
- LT Switch → Keep in Lead as it is

## Step 2 (Cab #2)

- Reverser → Insert → Move Forward/Reverse
- EC Switch → Isolate / Run (When moving)
- A9 → FS for 34 Sec then move to free
- SA9 → Move from Apply to Release
- Parking brake → Release (If applied)
- LT Switch → Lead



# Multiple Units Operation



1. Energize all Loco's which are to be connected in MU one by one by selecting appropriate cab's
2. Connect MU cables between Loco's
3. All necessary breakers and switches to be switched on in all the Loco's

| Lead Loco | Trail Loco |
|---|---|
| • Throttle → Idle<br>• Reverser → Centered<br>• LT Switch → Lead (Confirm from TCD display)<br>• Front side MU connector → Close with connector cap<br>• Circuit Breakers (Control, GFB & DB) → ON<br>• EC Switch → Isolate | • Throttle → Idle<br>• Reverser → Centered & removed<br>• LT Switch → Trail (confirm from TCD display)<br>• A9 → Keep in FS<br>• SA9 → Keep in 'Rel'<br>• Circuit Breakers (Control, GFB & DB) → ON<br>• EC Switch → Isolate |

**Power Requirement:**

- Both the Loco's → EC Switch position should be 'Run' in both the Loco's
- Only Lead Loco → EC switch of Lead Loco should be in 'Run' and Trail Loco should be in 'Isolate'
- Only Trail Loco → EC switch of Trail Loco should be in 'Run' and Lead Loco should be in 'Isolate'

**NOTE:**

1) If leading loco failed then clear section from Trailing loco with engine pushing → use G & SR
2) If there is fault in trailing Loco, it is known as Train Line Fault → Troubleshooting only from trailing Loco



# Cab Changing-Multiple Loco



Situation:
Loco pilot is driving the MU Loco's → Loco #1 in Lead & Loco #2 is in trail → Loco pilot wants to convert the Loco #2 as lead & Loco #1 as trail Loco

Follow the below steps for cab changeover

## Step 1 (Loco #1)

- SA9 → Apply
- Apply PB
- Remove Reverser

## Step 2 (Loco #2)

- LT Switch → Lead
- A9 → Run
- SA9 → Apply

## Step 3 (Loco #1)

- A9 → FS
- SA9 → Release
- LT Switch → Trail
- PB → Release

## Step 4 (Loco #2)

- SA9 → Release
- Insert Reverser & move forward
- Move the train



# Dead Loco



| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Apply Parking Brake | EC panel | Red push button |
| 02 | Release Independent Brake (SA9) | EAB panel – Working cab | Move SA9 to REL position |
| 03 | Apply Automatic Brake (A9) | EAB panel – Working cab | Move A9 handle to FULL position. Ensure brake cylinder value of 1.8 +/- 0.1 Kg/cm2 on TCD |
| 04 | Couple to Live or Running Locomotive | | Direct or though rail cars |
| 05 | Reverser → Center, L & T switch → Trail | EAB panel – Working cab | Verify Equalizing Reservoir pressure goes to 0 |
| 06 | Turn off Air brake circuit breakers | EC panel – 3rd row | ABCB1 and ABCB2 |
| 07 | EC switch → Off | Working cab | |
| 08 | Switch off all Circuit breakers & Battery Switch | EC panel | Shutdown the Locomotive |
| 09 | Move 'Dead switch' → DEAD position | Pneumatic Operating Unit → Rad cab | BCE part of POU |
| 10 | Press push button → MC 30 valve for 30 secs | Pneumatic Operating unit → Rad cab | To release any trapped air pressure |
| 11 | Drain MR1 and MR2 | Underframe / gear | Open drain valve at MR1 & MR2 & close the drain valves after MR1 & MR2 are completely drained |



# Dead Loco



| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 12 | Verify for no air exhaust from BCE pipe | Underframe | Open any one side of BCE pipe angle cock to verify |
| 13 | Connect the BP of dead locomotive → live / running locomotive & open both ends of brake pipe angle cocks between the two locomotives | Underframe | To ensure brake pipe continuity → This allows charging of main reservoir no. 2 from brake pipe up to a dead engine setting of 1.8 - 2.0 Kg/sq. cm |
| 14 | Recharge the brake pipe from live locomotive to 5.0 +/- 0.1 Kg/cm2 | | Or rail cars connected to a live locomotive |
| 15 | Ensure that both the bogies BC-1 & BC-2 cocks are in cut-in position | Underframe | |
| 16 | Release parking brakes of bogie-1 & 2 | Underframe | Manual Release. Visually verify that all the brake shoes are released |
| 17 | Test for proper operation of brake shoes on all the wheels before clearing for movement | Underframe | Reduce the pressure in the brake pipe and verify that brake shoes get applied on all the wheels. Charge the brake pipe and verify that brake shoes released on all the wheels |
| | | | |

# Backup brake operation

EAB console becomes inactive ? Air brake fault msg displays on TCD

| Step No | Operational Step | Location | Remarks |
|---|---|---|---|
| 01 | Keep Throttle Idle and Reverser Centered | Working cab | |
| 02 | On Pneumatic panel, keep toggle as per working cab | Radiator cab | |
| 03 | Change mode switch to 'Back up' on pneumatic panel | Radiator cab | |
| 04 | Move backup handle backwards for BP charge | Working cab | |
| 05 | Move backup handle front for BP destroy | Working cab | |
| 06 | Observe BP, BC, ER and MR reading on Analog gages | Working cab | 1st gage →BP(white) and BC (Red)<br>2nd gage →ER(white) and MR(Red) |
| 07 | Clear the section by moving the Loco at 10 kmph speed | | |





| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Engine | **Engine not cranking** – Status message on TCD screen displays like 'Engine can't crank' or 'Engine won't start' | Battery voltage may be less than 55 volts | Check the battery knife switch (BS). Open the BS & close it again. Check the battery voltage once again. If it is less than 55 volts, contact GEDL helpline number |
| | | Aux cab doors may be open | Check for proper closure of CA2, CA3, CA4 & CA5 doors |
| | | Some of the circuit breakers in the engine starting circuit might have tripped. | Check & ensure all circuit breakers in 3rd and 4th of breaker panel along with COB are 'ON'. (LCCB, BCCB, MTB, ABCB1, ABCB2, FPB etc) |
| | | EC switch may not be in correct position | Ensure EC switch is in 'Isolate' position in working cab / stand and in 'off' position in non working cab / stand. |
| | **Engine cranking but not starting** – Engine not picking up the speed or running continuously | Battery voltage may be less than 55 volts | Check the battery knife switch (BS). Open the BS & close it again. Check the battery voltage once again. If it is less than 55 volts, contact GEDL helpline number |
| | | Some of the circuit breakers or contactors in the engine starting circuit may not be working properly. | Check & ensure all circuit breakers in 3rd and 4th of breaker panel along with COB are 'ON'. (LCCB, BCCB, MTB, ABCB1, ABCB2, FPB etc) |
| | | EC switch may not be in correct position | Ensure EC switch is in 'Isolate' position in working cab / stand and in 'off' position in non working cab / stand. |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Engine | **Engine cranking but not starting** – Engine not picking up the speed or running continuously | The 'Reverser' handle may not be inserted in the 'Master Controller" of the active cab / stand | If not placed it properly, insert and push it inside the slot |
| | | Sufficient fuel is not available | Check the fuel level in the tank |
| | **Engine started but unable to Load** – Loss of tractive effort | Locomotive faults – Fault message details appearing on the TCD | Check the TCD screen for fault message which restricts the loading. Take appropriate actions to clear the fault. Take the help from GEDL helpline center whenever required |
| | | Parking brake not released | Check physically for the release of Parking brakes on all the wheels.<br>If the train is having "Multiple Units', check the PB status on rear locomotives also. |
| | | GFB circuit breaker might be tripped | Check the status of GFB breaker and turn it 'ON', if it is tripped |
| | | EC switch may not be on 'Run' position in active cab / stand & 'Off' position at inactive cab / stand | Check the EC switch status and ensure the switch is in 'Run' position at active cab / stand and 'Off' position at inactive cab / stand |
| | | A message may be appearing on TCD screen showing 'Engine warming in progress' | Engine is still not at operating temperature. Wait for some time for the message to disappear and 'Ready' message displayed |
| | | A message may be appearing on TCD screen showing 'AESS is active' | Wait for about 4 min after moving 'Reverser' forward or reverse before moving the Throttle |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Engine | **Trail units not loading** | EC switch may not be on 'Run' position in active cab / stand & 'Off' position at inactive cab / stand of trailing unit | Check the EC switch status and ensure the switch is in 'Run' position at active cab / stand and 'Off' position at inactive cab / stand |
| | | MU cable may be loose or disconnected during loco movement | Verify the MU cable / connection along with air pipe connections |
| | | EAB system in trail unit might not be set up properly | Check with EAB system in trail unit. L & T switch should be in 'Trail' position in working cab / stand of trailing loco. |
| | | Some of the circuit breaker of trailing loco might have tripped | Check all CB status and reset wherever required |
| | | Rear engine may be shutdown | Check with 'MU Shutdown' switch of lead loco and ensure it is in release condition<br>Check and ensure the rear engine is started and running |
| | **Engine derated or 'Tractive power restricted'** | Locomotive may be traveling in tunnel and engine may be hot (At 90.5 °C engine power starts derating and at 115.5 °C, engine shuts down | Clear the section with available power |
| | | Low 'Lube oil' pressure. Check for the message on TCD screen | • Ensure drain valves are closed.<br>• Check for any leakage in engine cab<br>• Check the engine oil level using dipstick method. Oil should be above 'ADD' mark when engine is running at Idle<br>• Communicate with GEDL helpline center with your findings & for further instructions |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Engine | **Engine derated or 'Tractive power restricted'** | Low 'Coolant' pressure. Check for the message on TCD screen | • Check for any leakage in engine / rad cab<br>• Check the water level in 'Spy or Sight' glass. Water level should be above 'Min' level when engine is running in Idle<br>• Communicate with GEDL helpline center with your findings & for further instructions |
| | | More than one 'Traction motor' might be cut out (Manual or automatic) | Reduce the load accordingly |
| | **Engine not cranking after AESS recovery**- When 'Reverser' put to forward after 4 minutes – no auto cranking | • If any changes made in between engine shutdown by AESS and restart, then engine will not crank on its own.<br>• Might have used AESS facility more than 8 times between 00 Hrs. to 00 Hrs. | • If any changes are done or AESS facility is used for 8 times in the day, then we should manually start through START button only<br>• Try by releasing A9 and SA9. If the trouble is repeated via AESS then bypass the AESS by switching off the toggle switch provided at CA1 near CIO of cab1.<br>• Report the issue to GEDL helpline center |
| | **Engine not taking Load** -"Load Limited Cold Engine" message is shown on the screen | Engine is in very cold condition or less than the minimum operating temperature | • Allow the engine to run for some time and heated up<br>• If problem continues to be there even after 15 to 20 min, contact GEDL helpline center |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Brake system | **EAB communication lost or Air brake fault message is displayed on screen** | BP drops automatically on run | • Stop the loco and check for leakage<br>• Recycle all the circuit breakers and reset the system<br>• Clear the section with 'Backup Brake' operation<br>• Contact GEDL helpline center with your findings & for further instructions |
| | **Parking brake not released** | • May be electrical connection problems or malfunctioning of electrical components<br>• Mechanical jam of PB system with wheel | • Press 'PB release' button 2 to 3 times and check for disappearing of (((P))) mark on TCD screen. Try with other cab also if required<br>• Check the PB condition on wheels 1, 5, 8 & 12. Manually release it with pull handle till the entire pressure is released<br>• Don't close PB cock with MR pressure<br>• Contact GEDL helpline center with your findings & for further instructions |
| | **Brake pipe (BP) pressure not building** | • Air compressor – Look for any message on TCD screen related to compressor<br>• BP angle cock opened<br>• ALP emergency valve might have operated<br>• Look for any penalty or leakages | • Check for the incident on screen & contact helpline center for further instructions<br>• Check all the cocks in air brake system<br>• Close ALP emergency valve & retry<br>• Recover the penalties if any. Check the piping/joints, Air dryer, Air filter drains, stuck relief valves etc. for any defects |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Brake system | **ER & BP not charging** | L&T switch may be placed at wrong position | • Check and ensure the L&T switch is in 'Lead' position<br>• If L&T switch is in other than 'Lead' position, first ensure EC switch is in 'Isolate', keep A9 in 'FS' then move L&T switch to 'Lead' position. After this move A9 to 'Run' position. |
| | **ER & BP not charging -** When loco arrives as 'Dead' loco | Pneumatic Operating Unit (POU) set up or L&T switch may not be in correct position | • Check & ensure correct status of 'Dead', 'Backup', 'Passenger/Goods' & 'toggle' switches<br>• Ensure L&T switch is in 'Lead' position<br>• If L&T switch is in other than 'Lead' position, first ensure EC switch is in 'Isolate', keep A9 in 'FS' then move L&T switch to 'Lead' position. After this move A9 to 'Run' position. |
| | **BP not charging** - After attaching the load | L&T switch may be in 'Test' position | • Keep A9 in 'FS', and EC switch in 'Isolate' position, change L&T switch position to 'Lead', release A9 to 'Run' position. Then check BP building |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Operation | **Locked Axle / Traction motor cut out** | • Wheel may be locked due to axle related issue.<br>• Sensor may be defective. | • Locomotive must be stopped<br>• Physically verify the affected / indicated wheel / axle<br>• Contact GEDL helpline center and cutout the TM & its speed sensor or bypass the blocked axle switch at screen<br>• When cutting out the Traction motor, also cutout its speed sensor |
| | **Isolation of Traction motor** | • Traction motor winding temperature is beyond the limit<br>• Defective Traction motor | Cutout the respective Traction motor which is having problems. Follow the below process<br>• Select Switches option on screen<br>• Select Motor cutout option. Motor cutout screen appears<br>• Use function keys and number keys to isolate TM's and speed sensors.<br>• Isolation of TM should be informed to GEDL helpline center |
| | "**Don't Shut Down the Loco**" - Message is shown on the screen | Locomotive related issue | • Move loco 2 to 3 time for a small distance, forward / reverse<br>• Keep 'Reverser' in F/R for 2 or 3 times and try.<br>• Wait for 5 min. for message to change<br>• If still problem exits, contact GEDL helpline center |

| Component | Nature of trouble | Probable reasons | Troubleshooting |
|---|---|---|---|
| Locomotive Operation | **TCD screen or keyboard are not responding** | All 4 screen of the active cab or any one of the screens are not responding - TCD screen issue | • Stop the locomotive<br>• Recycle the BCCB<br>• Open the EC panel and recycle the individual breakers of each TCD screen<br>• Contact GEDL helpline center with your findings & for further instructions |
| | | Monitors are blank – Screen saver mode may be active or one of the breaker is tripped | • Press any key on the keyboard to make the monitor active<br>• Check BCCB or individual breakers of each screen and ensure they are switched 'ON' |
| | **Horn not active or not audible** | • Horn switch may be defective<br>• Cock may be closed<br>• Compressor trouble message is displayed | • Use other horn on ALP side<br>• Check the cock near head light<br>• For compressor problem, contact GEDL helpline center |
| | **Continuous Alarm and 'Alarm silence' screen is not displayed** | Check the locomotive for locked axles or a dead engine | Contact GEDL helpline center |
| | **Continuous Alarm upon manual Engine shutdown** | Engine related issue | To silence the alarm, move EC switch to 'Isolate' position. Contact GEDL helpline center for further guidance |
| | **Trainline Alarm from a Trail Unit** | Trouble in rear loco | Stop the train and go to rear loco. Reset the trainline faults from rear loco |
| | **Alerter cut-off** - Not responding. Alerter acting and BP dropping on run when crew is alert | Defective Alerter sensing | • Reset the Alerter from the active cab only. It cannot be done from inactive cab.<br>• If Alerter facility to be bypassed contact GEDL helpline center |

| Item | ⚠ Caution | ⚠ Warning |
|---|---|---|
| Parking Brake | Illuminating or non-illuminating of red indicator light in operator cab does not guarantee that Parking Brakes are fully applied or released. These lights are only for indication. | • Pilot must verify physically all PB's for proper application or release before leaving or moving the train respectively. If the locomotives are in consists, this applies to all MU's in consist.<br>• Release or disable the Parking Brakes prior to maintenance |
| Air reservoir (MR tank charging) | For quick MR charging, do not increase the engine speed. CIO will be monitoring the MR charging and it will take appropriate action based on the requirement. | |
| Change over from motoring mode to dynamic brake mode and back to motoring mode | When changing over from motoring mode to dynamic brake mode or from dynamic brake mode to motoring mode, pause for 10 seconds with combined power handle at 'Setup' | |
| TM cutout | If there are any issues or faults with traction motor operation, CIO senses the condition and initiates for automatic traction motor cut out | For manual traction motor / sensor cutout operation, follow specific railroad operating procedures. It is recommended that traction motors be manually cutout only when the 'Throttle' handle is in Idle |
| Locked axle alarm cut out switch | Before enabling the locked axle alarm cutout switch, ensure the wheels are rolling and not locked up. The switch graphic will change after enabling the cutout switch. | |
| Dead loco movement | To avoid wheel flats when operating locomotive as 'Dead', drain the main reservoir (MR) of the unit to less than 1.78 kg/cm²(25 PSI). Failure to do may result in equipment damage | |

| Item | ⚠ Caution | ⚠ Warning |
|---|---|---|
| Moving the train | • When moving the train, wait for at least four minutes after moving the 'Reverser' handle forward or reverse and before moving the 'Throttle' handle. This pause allows units that may have been shutdown by AESS to start the diesel engine. Failure to do so may result in alarms & faults. It may prevent consists from loading<br>• Pause 2 to 3 seconds while moving throttle from one notch to another notch.<br>• Pause 2 to 3 seconds while moving throttle from 'Idle' to 'Setup'. Apply Dynamic<br>• Ensure 'TELS' switch is moved to 'LIMITED' while limiting the Tractive Effort. | If the BP pressure is below 3.2 kg/cm²(45 PSI), train should not be permitted to continue operation. Train must be stopped and BP to be recharged to 5 kg/cm²(70 PSI). Failure to comply with this warning may result in the inability to control or stop the train. |


Wabtec
CORPORATION

# Helpline contact number

# 1800 120 66 88



## Details Required By Helpline Center:

- Name, Designation, Mobile No., Loco No., Train No., Load.
- Operating cab, EAB position (lead/trail).
- Incident | Observation | Alerter | Alarm | Operator Msg.
- Condition of cab equipment's like A9, IAB, PB, Breakers, Cutout cocks, Drain cocks etc.